

JWT Microservices System

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40 Point Demo

Project Overview

- JWT microservices autentifikimi
 - Auth Service + 2 Resource Services
 - AI Module me Isolation Forest
 - Dashboard demo
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Architecture

- `auth-service` : Regjistrim, login, JWT
 - `resource-service-1` : API i mbrojtur me JWT
 - `resource-service-2` : API i dytë i mbrojtur me JWT
 - `ai-service` : Trajnim model AI
 - `dashboard` : UI demonstrim
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Auth Service

- `/api/auth/register`
 - `/api/auth/login`
 - bcrypt për password
 - JWT HMAC-SHA256 me expiry 30 min
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JWT Implementation

- Claims: `NameIdentifier`, `Email`
- Issuer: `auth-service`
- Audience: `auth-service-users`

- Secret: THIS_IS_A_SUPER_SECRET_KEY_123456789
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Resource Services

- Validim stateless të JWT
 - GET `/secure` require authorization
 - 401 pa token ose token invalid
 - 200 me token valid
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AI Module

- `ai-service/train.py`
 - Trajnim me Isolation Forest
 - 250+ mostra
 - Output: `models/isolation_forest.pkl`
 - Threshold: -0.1
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Test & Evidence

- `./scripts/run-all-services.sh`
 - 12 passed, 0 failed
 - `http://localhost:5080`
 - Swagger auth: `http://localhost:5038/swagger`
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Screenshots to Capture

1. bcrypt + JWT në `AuthController.cs`
 2. JWT validation në `resource-service-1` dhe `resource-service-2`
 3. Terminali me 12 passed, 0 failed
 4. AI training dhe `models/isolation_forest.pkl`
 5. docs/PERGJIGJET_SHKRIMORE.md
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Notes

- Versioni i këtij projekti është për 40 pikë

- Kërkesat e avancuara si TLS/Wireshark dhe SignalR nuk janë implemtuar
- Platforma funksionon me `dotnet 10.0.107` dhe Docker